Test -3

14/3/20

1) GIT:

Git is a version control system used to track changes in code.

It helps developers manage project history and collaborate with others.

2) Github:

Github is a cloud platform that stores Git respositories online.

It allows developers to:

* Share code
* collaborate with others
* Manage projects.

3)

Git	Github
* Version control tool	* cloud hosting service
* Installed on local system	* Works on internet
* Tracks code changes	* Store Git repositories
* Command line tool	* web platform

4.) Staging Area:

The staging area is a place where changes are prepared before committing.

Steps:

1) Modify files
2) Add to staging area
3) Commit changes

Example:

```
git add file.html
```

5) Commit in Git:

A commit saves a snapshot of the project at a specific time.

```
git commit -m "update all"
```

6) Branch:

A branch is a separate version of the project used to develop features without affecting the main code.

command-line:

```
git branch feature-login
```

7)

git pull	git clone
* updates existing repository.	* Downloads repository for first time.
* Fetches latest changes	* creates local copy
* Used regularly	* used once.
command:	command:
git pull origin main	git clone repository-url

8) git add.

This command adds all modified files to the staging area.

. means current folder files.

9) Use of git status:

It shows :- Modified files
- staged files
- Untracked files

10) Origin in Git :

Origin is the default name of the remote repository.

CSS

1) CSS & its use :

CSS (cascading style sheets) is used to style HTML elements.

It controls :

Colors Fonts Animations
Layout spacing

Example :

```
P{
 color: red;
}
```

This makes paragraph text red.

2) Three types of CSS :

a) Inline CSS

Applied inside HTML tag

```
<html>
  <head> </head>
  <body>
    <p style = "color: pink;" > Hello</p>
  </body>
</html>
```

b) Internal CSS

Written inside < style > tag in < head >

Example:

```
< style >
    p {
        color: blue;
    }
</ style >
```

c) External CSS

written in a separate .css file.

HTML:

```
< link rel = "stylesheet" href = "style.css" >
```

CSS:

```
p {
    color: green;
}
```

3) CSS selector:

A selector selects HTML elements to apply styles.

Types of selectors:

1) Element selector p { }
2) Class selector . className { }
3) ID selector # id Name { }
4) Universal selector * { }

4) CSS called Cascading Style sheets why?

Because multiple styles can apply to the same element, and CSS decides which one should apply based on priority rules.

Example :

```
P{
color: blue;
font-size : 10px;
}

P {
color: red; }
```

This last rule overrides the previous one.

```
.idname {
display: flex;
flex-direction: column;
}
```

5) Specificity in CSS :

Specificity determines which CSS rule has higher priority.

Inline → ID → class → Element

ID has the highest priority.

6) CSS Box Model :

The box model defines layout and spacing of elements.

Content → padding → Border → Margin.

Example :

```
div{
  padding: 20px;
  border : 2px solid black;
  margin : 30px;
}
```

7)

Margin	Padding
* Space outside element	* Space inside element
* Separates elements	* Adds space inside box
Example : margin: 20px;	Example : padding: 20px;

8) Flexbox :

Flexbox is a layout system used to align elements in rows or columns

Used for :

* Centering
* Responsive layouts.
* Equal spacing.

Example :

```
. container {
  display: flex;
  justify-content : center;
  align-items : center;
}
```

9) CSS Grid:

CSS grid is a layout system for two-dimensional layout (rows + columns).

Example:

```
.container {
    display: grid;
    grid-template-columns: 1fr 1fr 1fr;
}
```

Flexbox	Grid
* One-direction (row/column)	* Two directions (row + column)
* Good for components	* Good for page layouts

10) 1fr mean in Grid:

fr = fraction of available space

Example:

```
grid-template-columns: 1fr 1fr 1fr;
```

Each column take equal space.

repeat ()

used to repeat columns

Example:

```
grid-template-columns: repeat (3, 1fr);
```

11) Position Property Values:

* Static → Default
* relative → Moves relative to original position
* Absolute → Positioned relative to parent
* fixed → Stays fixed on screen
* sticky → Switches between relative and fixed

Example:

```
.box {
  position: absolute;
  top: 20px;
  left: 50px;
}
```

12) z-Index:

z-index controls stacking order of elements.

Higher value appears on top.

Example:

```
.box 1 {
  z-index: 1;
}
.box 2 {
  z-index: 10;
}
```

box 2 will appear above.

13)

Pseudo-class	Pseudo-element
* Style element state	* Styles part of element
* Uses single :	* Uses ::
Examples :	
button : hover {	p :: first-letter {
background : red;	font-size : 30px;
}	color : red;
	}

14)

Transform	Transition
* changes shape or position	Adds animation between states
transform : scale (1.2);	transition : 0.5s;

15) Media Query :

A media query applies CSS based on screen size.

Used for responsive design.

```
@ media (max-width : 768px) {
    .container {
        flex-direction : column;
    }
}
```